PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 11 · Simulations

THURSDAY NOV 5, 2026

01

PULLING THE GOALIE

YOU'RE DOWN ONE GOAL LATE IN THE THIRD PERIOD. WHEN DO YOU PULL YOUR GOALIE?

MONTE CARLO SIMULATION

A model of uncertainty in which you run many random trials to estimate the likelihood of various outcomes.

The technique was formalized during the Manhattan Project to model reactions that were too complex for analytic equations.

STRENGTHS AND LIMITATIONS

STRENGTHS

Allows for complex problems to be solved without advanced math.

Shows a full range of outcomes.

LIMITATIONS

Reliant on probability inputs.

Often assumes independence of events you are simulating.

Can become computationally intensive.



WHAT WOULD YOU NEED TO KNOW TO SIMULATE THE END OF A GAME WITH THE GOALIE PULLED?

? WHAT MIGHT OUR GOALIE MODEL HAVE MISSED?

02

TWO STREAKS

? WHICH RECORD IS MORE IMPRESSIVE: JOE DIMAGGIO'S 56-GAME HITTING STREAK, OR CAL RIPKEN'S 2,632 CONSECUTIVE GAMES PLAYED?

THE INPUTS

	RATE	CHANCE IN ANY ONE GAME
DiMaggio, 1941	.357 average, 3.9 at bats per game	82% chance of a hit
Typical regular, 1936 to 1946	.287 average, 3.8 at bats per game	72% chance of a hit
Everyday players, 1982 to 1998	Share of team games played	91% chance of playing

Source: Lahman Baseball Database (R package Lahman 14.0).



WAS RIPKEN'S RECORD THE MORE IMPRESSIVE ONE? WHY OR WHY NOT?

03

SIMULATING A SEASON

2025 NFL, FROM WEEK 9

1. Rate each team, adjusting for schedule.
2. Turn ratings into a win probability. Consider home field advantage and standard deviation.
3. Simulate the 137 remaining games.
4. Seed the playoffs.
5. Simulate the bracket.
6. Summarize the results.

Source: nflverse games, 2025 season. Ratings and odds as of the end of Week 9.

PLAYOFF ODDS AFTER WEEK 9, 2025

TEAM	RECORD	AVG MARGIN	RATING	PLAYOFFS	SUPER BOWL
PHI	6–2	+2.9	+5.0	99%	5%
SEA	6–2	+10.1	+11.2	99%	20%
TB	6–2	+2.4	+4.9	99%	5%
LA	6–2	+10.2	+11.4	99%	20%
NE	7–2	+7.6	+2.0	98%	3%
DEN	7–2	+6.6	+5.6	97%	7%
BUF	6–2	+8.5	+4.8	95%	5%
IND	7–2	+12.1	+10.1	95%	19%
DET	5–3	+7.6	+6.3	89%	4%
SF	6–3	+0.7	+4.1	89%	2%

Source: nflverse games, 2025 season. Ratings and odds as of the end of Week 9.



WHAT WAS OUR MODEL MISSING?

ACCOUNTING FOR RATING ERROR

1 **This season's noise.**

A team's margin swings about 13 points a game, so after 8 games its rating is off by about 4.4 points. To adjust for this, you can keep 54% of each team's rating and regress to the mean.

2 **Prior seasons.**

You can also look at prior seasons to assess stability. From 2015 to 2024, the rest-of-season rating kept 59% of the Week 9 rating, and real strength moved about 3.8 points either way after Week 9.

3 **Either way you measure this instability,**

draw each team's rating fresh in every simulated season around its pulled-in rating. Both approaches raise the expected misses among the 10 favorites from 0.4 to 1.2.

PLAYOFF ODDS WITH RATING ERROR

TEAM	NO ERROR	THIS SEASON	PRIOR SEASONS
PHI	99%	94%	93%
SEA	99%	92%	92%
TB	99%	92%	92%
LA	99%	89%	89%
NE	98%	94%	94%
DEN	97%	93%	93%
BUF	95%	89%	89%
IND	95%	92%	92%
DET	89%	71%	72%
SF	89%	78%	78%

Source: nflverse games, 2025 season. Ratings and odds as of the end of Week 9.

04

THIS WEEK'S WORK

CASE STUDY

SIMULATING A SEASON

STEPS

1. Pick a sport from weeks/week11/data and load its played and remaining games.
2. Rate every team and turn ratings into win chances. Check them on games already played.
3. Simulate the rest of the season and the postseason 10,000 times.
4. Report each team's odds, and build a page with a slider that changes one input.
5. Commit your predictions.

BRIEF QUESTIONS

1. What assumption in your model do you trust least? How does that assumption impact your odds?
2. Was there a team your model liked more than you expected? Why do you think your model liked that team more than you would've anticipated?

Written to a beat writer covering the team you care most about.

FINAL PROJECT PROPOSAL

1. Research question.
2. Potential strategies for answering the question.
3. Identified data sources. Be specific.
4. Expected challenges.

BEFORE YOU PUSH ON MONDAY, NOVEMBER 9

Commit your code, your odds table, your interactive page and predictions.csv.

Answer every question in BRIEF.md, in your own words.

Run /audit, then /quiz-me.

Upload your repository's link to Canvas by Monday, November 9 at 11:59pm.